CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

CORREIA FAMILY DAIRY

26380 W FAHEY RD

GUSTINE CA 95322

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Correia Family Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
26380 W Fahey RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1984

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X020-X270-X008-XXXX

B. OPERATORS

Correia, Ezquiel Jr

Operator name: Correia, Ezquiel Jr

Telephone no.: Landline: ; Cellular: (209) 652-0029

Mailing Address Number and Street	City	State	Zip Code
23680 W Fahey RD	Gustine	CA	95322

C. OWNERS

Correia, Ezequiel

Legal owner name: Correia, Ezequiel

Telephone no.: Landline: (209) 826-1903; Cellular:

Mailing Address Number and Street	City	State	Zip Code
26380 W Fahey RD	Gustine	CA	95322

This owner is responsible for paying permit fees.

Correia, Ezquiel Jr

Legal owner name: Correia, Ezquiel Jr

Telephone no.: Landline: ; Cellular: (209) 652-0029

Mailing Address Number and Street	City	State	Zip Code
23680 W Fahey RD	Gustine	CA	95322

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	332	283	213	129
Number under roof	1,144	167	0	0	0	0
Maximum number	1,144	167	332	283	213	129
Average number	1,144	167	332	283	213	129
Avg live weight (lbs)	1,200	1,250	875	550		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 69 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 38,576.24 *tons per reporting period*

Total nitrogen from manure: 488,236.61 *lbs per reporting period*

After ammonia losses (30% loss applied): 341,765.63 *lbs per reporting period*

Total phosphorus from manure: 79,766.82 *lbs per reporting period*

Total potassium from manure: 218,334.95 *lbs per reporting period*

Total salt from manure: 577,054.05 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,100,000 *gallons*

Total nitrogen generated: 80,492.13 *lbs*

Total phosphorus generated: 15,179.55 *lbs*

Total potassium generated: 135,714.11 *lbs*

Total salt generated: 709,619.58 *lbs*

	23,100,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	23,100,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID water	Surface water
Domestic Well	Ground water
Drain water	Surface water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	Compost	5,096.00 *ton*	Dry-weight	40.8		18,200.00	5,200.00	20,900.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	109,812.68	31,375.05	126,103.58	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	109,812.68	31,375.05	126,103.58	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1-1	7	7	2	process wastewater	X070-X270-X008-XXXX
1-2	12	12	2	process wastewater	X070-X270-X008-XXXX
1-3	80	80	4	process wastewater	X070-X270-X014-XXXX
Frontella	40	40	2	both	X070-X270-X046-XXXX
Pavao	85	85	2	process wastewater	X070-X270-X110-XXXX X070-X270-X180-XXXX X070-X270-X190-XXXX X070-X270-X200-XXXX
Pavao pasture	38	38	0	none	X070-X111-X001-XXXX X070-X111-X002-XXXX
Pereira	75	75	0	none	X070-X270-X006-XXXX X070-X270-X021-XXXX
Vieira	63	63	2	manure	X070-X010-X014-XXXX
Totals for areas that were used for application	287	287	14		
Totals for areas that were not used for application	113	113	0		
Land application area totals	400	400	14		

B. CROPS AND HARVESTS

1-1

Field name: 1-1

09/25/2022: Ryegrass silage

Crop: Ryegrass silage Acres planted: 7 Plant date: 09/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	35.20 *ton*	Dry-weight		68.6	25,000.00	5,200.00	46,900.00		14.80
05/20/2023	35.20 *ton*	Dry-weight		68.6	25,000.00	5,200.00	46,900.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	10.06	157.90	32.84	296.22	934.75

1-2

Field name: 1-2

09/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 12 Plant date: 09/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	153.60 *ton*	Dry-weight		51.3	17,200.00	2,200.00	14,400.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	12.80	214.44	27.43	179.53	1,084.65

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	276.00 *ton*	Dry-weight		71.1	16,100.00	2,100.00	24,500.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.00	214.03	27.92	325.70	1,023.64

1-3

Field name: 1-3

1-3

09/25/2022: Ryegrass Silage

Crop: Ryegrass Silage Acres planted: 80 Plant date: 09/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	272.00 *ton*	Dry-weight		64.4	22,200.00	4,200.00	40,300.00		12.50
05/20/2023	398.00 *ton*	Dry-weight		79.0	32,700.00	5,000.00	43,000.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	8.38	122.07	20.61	187.41	588.86

05/28/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 80 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/15/2023	688.00 *ton*	Dry-weight		73.1	28,300.00	3,900.00	38,900.00		17.80
10/06/2023	592.00 *ton*	Dry-weight		73.1	28,300.00	3,900.00	38,900.00		17.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	16.00	243.61	33.57	334.85	1,532.22

Frontella

Field name: Frontella

Frontella

09/25/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 40 Plant date: 09/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	342.00 *ton*	Dry-weight		47.3	16,800.00	2,400.00	15,900.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	8.55	151.40	21.63	143.29	612.80

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	892.00 *ton*	Dry-weight		59.2	10,000.00	1,800.00	9,800.00		3.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.30	181.97	32.75	178.33	709.68

Pavao

Field name: Pavao

09/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 85 Plant date: 09/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	1,019.30 *ton*	Dry-weight		53.5	19,000.00	2,600.00	16,500.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	11.99	211.89	29.00	184.01	802.97

Pavao

05/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 85 Plant date: 05/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	2,023.00 *ton*	Dry-weight		65.0	12,600.00	2,200.00	15,200.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.80	209.92	36.65	253.23	882.98

Pavao pasture

Field name: Pavao pasture

12/01/2014: Pasture

Crop: Pasture Acres planted: 38 Plant date: 12/01/2014

No harvests entered for this crop.

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.00	60.00	2.00	2.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Pereira

Field name: Pereira

11/30/2018: Pasture

Crop: Pasture Acres planted: 75 Plant date: 11/30/2018

No harvests entered for this crop.

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.00	60.00	2.00	2.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Vieira

Field name: Vieira

09/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 63 Plant date: 09/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	696.00 *ton*	Dry-weight		53.1	11,800.00	2,000.00	13,900.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	11.05	122.28	20.73	144.04	911.91

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	1,587.60 *ton*	Dry-weight		67.4	9,200.00	2,600.00	11,000.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.20	151.16	42.72	180.73	887.24

NUTRIENT BUDGET

A. LAND APPLICATIONS

1-1 - 09/25/2022: Ryegrass silage

Field name: 1-1

Crop: Ryegrass silage Plant date: 09/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	119.29	12.84	166.23	1,026.44	210,000.00 *gal*
Irrigation Well	Ground water	3.50	0.00	0.00	0.00	840,000.00 *gal*
Application event totals		122.80	12.84	166.23	1,026.44	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.65	6.42	83.12	513.22	105,000.00 *gal*
Irrigation Well	Ground water	3.61	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		63.25	6.42	83.12	513.22	

1-2 - 09/27/2022: Wheat, silage, soft dough

Field name: 1-2

Crop: Wheat, silage, soft dough Plant date: 09/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	124.10	19.97	261.77	1,548.00	525,000.00 *gal*
Irrigation Well	Ground water	3.94	0.00	0.00	0.00	1,620,000.00 *gal*
Application event totals		128.04	19.97	261.77	1,548.00	

1-2 - 09/27/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	139.17	14.98	193.94	1,197.51	420,000.00 *gal*
Irrigation Well	Ground water	3.65	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		142.82	14.98	193.94	1,197.51	

1-2 - 05/01/2023: Corn, silage

Field name: 1-2

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	125.01	22.05	203.28	1,308.50	420,000.00 *gal*
CCID water	Surface water	5.28	0.00	0.00	620.98	4,464,830.00 *gal*
Application event totals		130.29	22.05	203.28	1,929.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.15	17.09	106.32	338.81	210,000.00 *gal*
CCID water	Surface water	5.20	0.00	0.00	611.92	4,399,650.00 *gal*
Application event totals		67.35	17.09	106.32	950.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	3.85	0.00	0.00	453.27	3,259,000.00 *gal*
Application event totals		3.85	0.00	0.00	453.27	

1-2 - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.15	17.09	106.32	338.81	210,000.00 *gal*
CCID water	Surface water	3.89	0.00	0.00	457.81	3,291,590.00 *gal*
Application event totals		66.04	17.09	106.32	796.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.74	0.00	0.00	321.82	2,313,890.00 *gal*
Application event totals		2.74	0.00	0.00	321.82	

1-3 - 09/25/2022: Ryegrass Silage

Field name: 1-3

Crop: Ryegrass Silage

Plant date: 09/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.35	14.38	188.48	1,114.56	2,520,000.00 *gal*
Irrigation Well	Ground water	6.45	0.00	0.00	0.00	7,110,000.00 *gal*
Application event totals		95.80	14.38	188.48	1,114.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.75	4.50	58.18	359.25	840,000.00 *gal*
Irrigation Well	Ground water	6.78	0.00	0.00	0.00	7,470,000.00 *gal*
Application event totals		48.53	4.50	58.18	359.25	

1-3 - 09/25/2022: Ryegrass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.02	0.00	0.00	0.00	7,740,000.00 *gal*
Application event totals		7.02	0.00	0.00	0.00	

04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.11	0.00	0.00	0.00	7,830,000.00 *gal*
Application event totals		7.11	0.00	0.00	0.00	

1-3 - 05/28/2023: Sudangrass, silage

Field name: 1-3

Crop: Sudangrass, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	13.48	0.00	0.00	0.00	14,850,000.00 *gal*
Application event totals		13.48	0.00	0.00	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.76	16.54	152.46	981.37	2,100,000.00 *gal*
Irrigation Well	Ground water	6.78	0.00	0.00	0.00	7,470,000.00 *gal*
Drain water	Surface water	0.62	0.00	0.00	0.00	7,470,000.00 *gal*
Application event totals		101.16	16.54	152.46	981.37	

1-3 - 05/28/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.55	28.19	175.42	559.03	2,310,000.00 *gal*
Irrigation Well	Ground water	8.00	0.00	0.00	0.00	8,820,000.00 *gal*
Drain water	Surface water	0.74	0.00	0.00	0.00	8,820,000.00 *gal*
Application event totals		111.29	28.19	175.42	559.03	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.68	0.00	0.00	0.00	12,870,000.00 *gal*
Application event totals		11.68	0.00	0.00	0.00	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.94	15.38	95.68	304.93	1,260,000.00 *gal*
Irrigation Well	Ground water	11.68	0.00	0.00	0.00	12,870,000.00 *gal*
Application event totals		67.62	15.38	95.68	304.93	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.68	0.00	0.00	0.00	12,870,000.00 *gal*
Application event totals		11.68	0.00	0.00	0.00	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.78	0.00	0.00	0.00	7,470,000.00 *gal*
Drain water	Surface water	0.62	0.00	0.00	0.00	7,470,000.00 *gal*
Application event totals		7.40	0.00	0.00	0.00	

Frontella - 09/25/2022: Wheat, silage, soft dough

Frontella - 09/25/2022: Wheat, silage, soft dough

Field name: Frontella

Crop: Wheat, silage, soft dough

Plant date: 09/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	135.95	25.43	33.73	0.00	368.00 *ton*
Application event totals		135.95	25.43	33.73	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	3.07	0.00	0.00	0.00	4,200,000.00 *gal*
Application event totals		3.07	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.63	6.74	87.27	538.88	630,000.00 *gal*
Irrigation Well	Ground water	3.24	0.00	0.00	0.00	4,440,000.00 *gal*
Application event totals		65.87	6.74	87.27	538.88	

Frontella - 05/30/2023: Corn, silage

Field name: Frontella

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.76	16.54	152.46	981.37	1,050,000.00 *gal*
CCID water	Surface water	3.01	0.00	0.00	353.55	8,473,400.00 *gal*
Application event totals		96.76	16.54	152.46	1,334.92	

Frontella - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	129.29	36.94	148.47	0.00	240.00 *ton*
Application event totals		129.29	36.94	148.47	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.06	0.00	0.00	242.73	5,817,315.00 *gal*
Application event totals		2.06	0.00	0.00	242.73	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.69	0.00	0.00	198.53	4,758,140.00 *gal*
Application event totals		1.69	0.00	0.00	198.53	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.45	0.00	0.00	288.28	6,909,080.00 *gal*
Application event totals		2.45	0.00	0.00	288.28	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.09	0.00	0.00	246.13	5,898,790.00 *gal*
Application event totals		2.09	0.00	0.00	246.13	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.62	0.00	0.00	308.68	7,397,930.00 *gal*
Application event totals		2.62	0.00	0.00	308.68	

Frontella - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.58	0.00	0.00	303.24	7,267,570.00 *gal*
Application event totals		2.58	0.00	0.00	303.24	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.41	0.00	0.00	283.52	6,795,015.00 *gal*
Application event totals		2.41	0.00	0.00	283.52	

Pavao - 09/27/2022: Wheat, silage, soft dough

Field name: Pavao

Crop: Wheat, silage, soft dough

Plant date: 09/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.08	11.28	147.82	874.16	2,100,000.00 *gal*
Irrigation Well	Ground water	11.22	0.00	0.00	0.00	13,140,000.00 *gal*
Application event totals		81.30	11.28	147.82	874.16	

03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	117.89	12.69	164.28	1,014.36	2,520,000.00 *gal*
Irrigation Well	Ground water	9.38	0.00	0.00	0.00	10,980,000.00 *gal*
Application event totals		127.27	12.69	164.28	1,014.36	

Pavao - 09/27/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.30	6.23	57.40	369.46	840,000.00 *gal*
Irrigation Well	Ground water	9.15	0.00	0.00	0.00	10,710,000.00 *gal*
Application event totals		44.44	6.23	57.40	369.46	

Pavao - 05/01/2023: Corn, silage

Field name: Pavao

Crop: Corn, silage

Plant date: 05/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.53	0.00	0.00	0.00	8,820,000.00 *gal*
Drain water	Surface water	0.69	0.00	0.00	0.00	8,820,000.00 *gal*
Application event totals		8.23	0.00	0.00	0.00	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	13.07	0.00	0.00	0.00	15,300,000.00 *gal*
Application event totals		13.07	0.00	0.00	0.00	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.59	12.45	114.80	738.92	1,680,000.00 *gal*
Irrigation Well	Ground water	6.30	0.00	0.00	0.00	7,380,000.00 *gal*
Drain water	Surface water	0.58	0.00	0.00	0.00	7,380,000.00 *gal*
Application event totals		77.48	12.45	114.80	738.92	

Pavao - 05/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.46	0.00	0.00	0.00	8,730,000.00 *gal*
Drain water	Surface water	0.69	0.00	0.00	0.00	8,730,000.00 *gal*
Application event totals		8.14	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.97	21.71	135.08	430.48	1,890,000.00 *gal*
Irrigation Well	Ground water	11.15	0.00	0.00	0.00	13,050,000.00 *gal*
Application event totals		90.12	21.71	135.08	430.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.15	0.00	0.00	0.00	13,050,000.00 *gal*
Application event totals		11.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	52.65	14.47	90.06	286.99	1,260,000.00 *gal*
Irrigation Well	Ground water	11.15	0.00	0.00	0.00	13,050,000.00 *gal*
Application event totals		63.79	14.47	90.06	286.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.23	0.00	0.00	0.00	7,290,000.00 *gal*
Drain water	Surface water	0.57	0.00	0.00	0.00	7,290,000.00 *gal*
Application event totals		6.80	0.00	0.00	0.00	

Pavao pasture - 12/01/2014: Pasture

Pavao pasture - 12/01/2014: Pasture

Field name: Pavao pasture

Crop: Pasture

Plant date: 12/01/2014

No nutrient budget entered for this crop.

Pereira - 11/30/2018: Pasture

Field name: Pereira

Crop: Pasture

Plant date: 11/30/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	0.92	0.00	0.00	108.79	4,888,500.00 *gal*
Application event totals		0.92	0.00	0.00	108.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.37	0.00	0.00	161.73	7,267,570.00 *gal*
Application event totals		1.37	0.00	0.00	161.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.15	0.00	0.00	134.89	6,061,740.00 *gal*
Application event totals		1.15	0.00	0.00	134.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.10	0.00	0.00	129.09	5,801,020.00 *gal*
Application event totals		1.10	0.00	0.00	129.09	

Pereira - 11/30/2018: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.04	0.00	0.00	122.56	5,507,710.00 *gal*
Application event totals		1.04	0.00	0.00	122.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	0.92	0.00	0.00	108.06	4,855,910.00 *gal*
Application event totals		0.92	0.00	0.00	108.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.10	0.00	0.00	129.82	5,833,610.00 *gal*
Application event totals		1.10	0.00	0.00	129.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	1.07	0.00	0.00	125.47	5,638,070.00 *gal*
Application event totals		1.07	0.00	0.00	125.47	

Vieira - 09/27/2022: Wheat, silage, soft dough

Field name: Vieira

Crop: Wheat, silage, soft dough

Plant date: 09/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	147.77	27.64	36.66	0.00	630.00 *ton*
Application event totals		147.77	27.64	36.66	0.00	

Vieira - 09/27/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	3.49	0.00	0.00	410.97	15,512,840.00 *gal*
Application event totals		3.49	0.00	0.00	410.97	

Vieira - 05/30/2023: Corn, silage

Field name: Vieira

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Compost	172.39	49.25	197.96	0.00	504.00 *ton*
Application event totals		172.39	49.25	197.96	0.00	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.87	0.00	0.00	337.58	12,742,690.00 *gal*
Application event totals		2.87	0.00	0.00	337.58	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.98	0.00	0.00	350.53	13,231,540.00 *gal*
Application event totals		2.98	0.00	0.00	350.53	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.55	0.00	0.00	299.59	11,308,730.00 *gal*
Application event totals		2.55	0.00	0.00	299.59	

Vieira - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.58	0.00	0.00	303.91	11,471,680.00 *gal*
Application event totals		2.58	0.00	0.00	303.91	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.55	0.00	0.00	299.59	11,308,730.00 *gal*
Application event totals		2.55	0.00	0.00	299.59	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.62	0.00	0.00	308.23	11,634,630.00 *gal*
Application event totals		2.62	0.00	0.00	308.23	

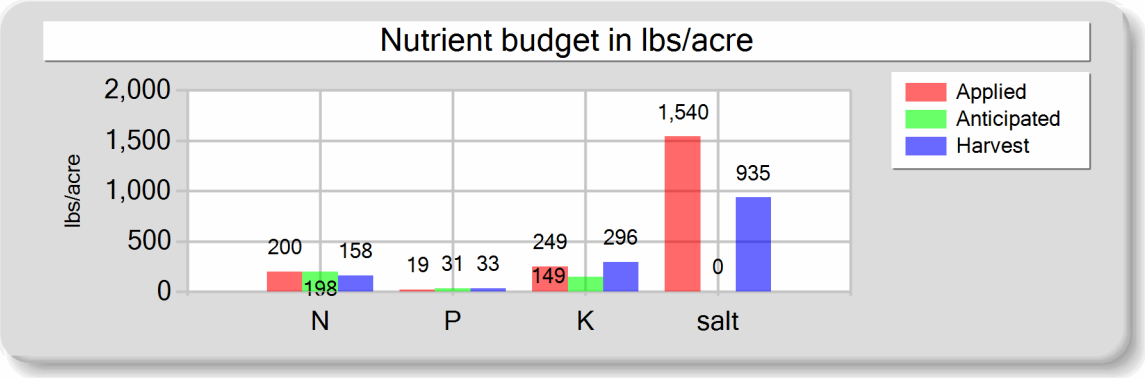
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID water	Surface water	2.11	0.00	0.00	247.79	9,353,330.00 *gal*
Application event totals		2.11	0.00	0.00	247.79	

B. NUTRIENT BUDGET

1-1 - 09/25/2022: Ryegrass silage

Field name: 1-1 Crop: Ryegrass silage Plant date: 09/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.94	19.26	249.35	1,539.65
Fresh water	7.11	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	200.05	19.26	249.35	1,539.65
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	157.90	32.84	296.22	934.75
Nutrient balance	42.15	-13.58	-46.87	604.90
Applied to removed ratio	1.27	0.59	0.84	1.65

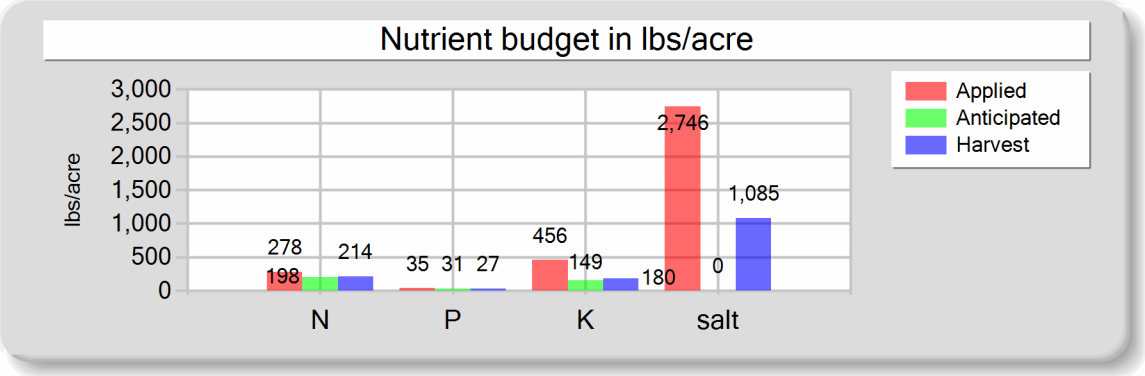
Fresh water applied	
1,704,000.00	*gallons*
62.75	*acre-inches*
8.96	*inches/acre*

Process wastewater applied	
315,000.00	*gallons*
11.60	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
2	*harvests*

1-2 - 09/27/2022: Wheat, silage, soft dough

Field name: 1-2 Crop: Wheat, silage, soft dough Plant date: 09/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.27	34.95	455.71	2,745.51
Fresh water	7.59	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.86	34.95	455.71	2,745.51
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	214.44	27.43	179.53	1,084.65
Nutrient balance	63.43	7.53	276.18	1,660.86
Applied to removed ratio	1.30	1.27	2.54	2.53

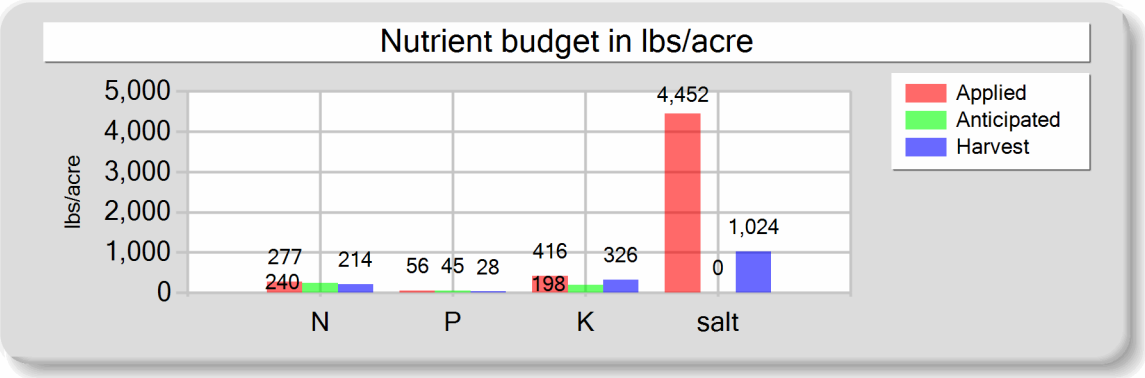
Fresh water applied	
3,120,000.00	*gallons*
114.90	*acre-inches*
9.57	*inches/acre*

Process wastewater applied	
945,000.00	*gallons*
34.80	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

1-2 - 05/01/2023: Corn, silage

Field name: 1-2 Crop: Corn, silage Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	249.32	56.22	415.91	1,986.11
Fresh water	20.96	0.00	0.00	2,465.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.27	56.22	415.91	4,451.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	214.03	27.92	325.70	1,023.64
Nutrient balance	63.24	28.31	90.21	3,428.27
Applied to removed ratio	1.30	2.01	1.28	4.35

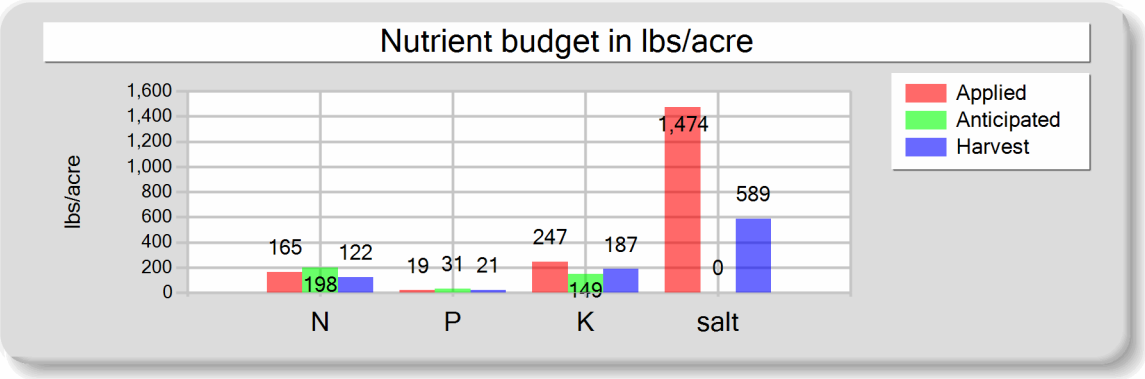
Fresh water applied	
17,728,960.00	*gallons*
652.90	*acre-inches*
54.41	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
2.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

1-3 - 09/25/2022: Ryegrass Silage

Field name: 1-3 Crop: Ryegrass Silage Plant date: 09/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	131.10	18.87	246.66	1,473.81
Fresh water	27.36	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.46	18.87	246.66	1,473.81
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	122.07	20.61	187.41	588.86
Nutrient balance	43.39	-1.74	59.25	884.95
Applied to removed ratio	1.36	0.92	1.32	2.50

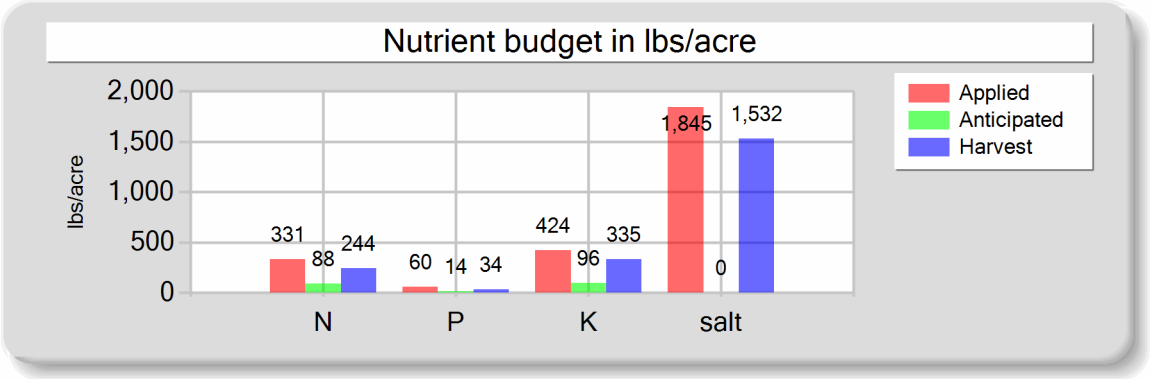
Fresh water applied	
30,150,000.00	*gallons*
1,110.32	*acre-inches*
13.88	*inches/acre*

Process wastewater applied	
3,360,000.00	*gallons*
123.74	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
2	*harvests*

1-3 - 05/28/2023: Sudangrass, silage

Field name: 1-3 Crop: Sudangrass, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.25	60.11	423.57	1,845.33
Fresh water	72.06	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.31	60.11	423.57	1,845.33
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	243.61	33.57	334.85	1,532.22
Nutrient balance	87.70	26.54	88.72	313.11
Applied to removed ratio	1.36	1.79	1.26	1.20

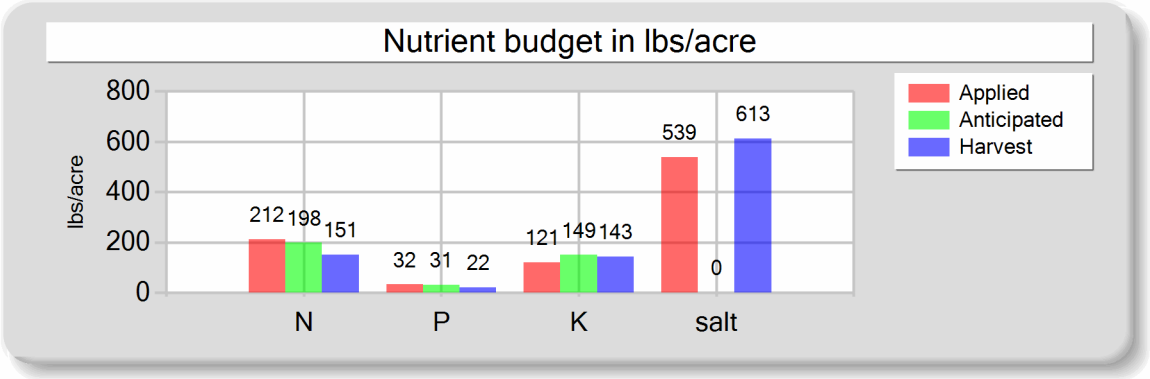
Fresh water applied	
100,980,000.00	*gallons*
3,718.75	*acre-inches*
46.48	*inches/acre*

Process wastewater applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
2	*harvests*

Frontella - 09/25/2022: Wheat, silage, soft dough

Field name: Frontella | Crop: Wheat, silage, soft dough | Plant date: 09/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	135.95	25.43	33.73	0.00
Process wastewater	62.63	6.74	87.27	538.88
Fresh water	6.31	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.88	32.17	121.00	538.88
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	151.40	21.63	143.29	612.80
Nutrient balance	60.49	10.54	-22.29	-73.92
Applied to removed ratio	1.40	1.49	0.84	0.88

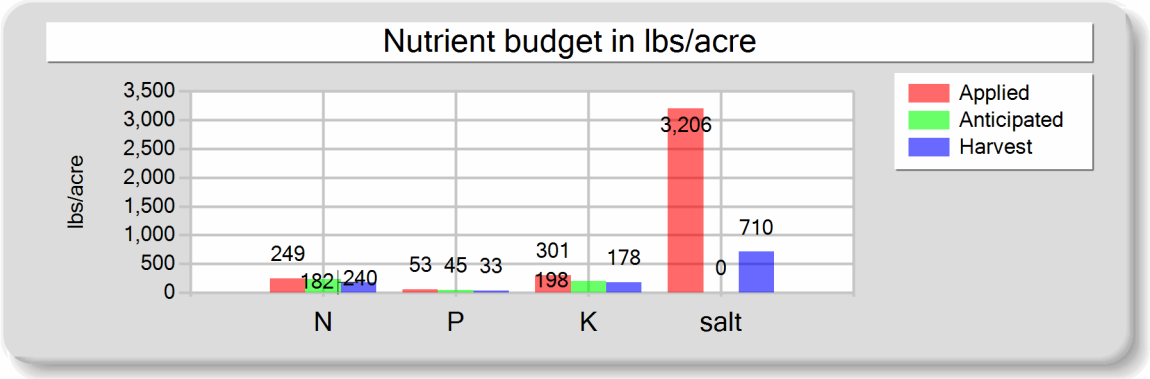
Fresh water applied	
8,640,000.00	*gallons*
318.18	*acre-inches*
7.95	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
0.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

Frontella - 05/30/2023: Corn, silage

Field name: Frontella Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.29	36.94	148.47	0.00
Process wastewater	93.76	16.54	152.46	981.37
Fresh water	18.91	0.00	0.00	2,224.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.96	53.48	300.94	3,206.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	181.97	32.75	178.33	709.68
Nutrient balance	66.99	20.73	122.61	2,496.36
Applied to removed ratio	1.37	1.63	1.69	4.52

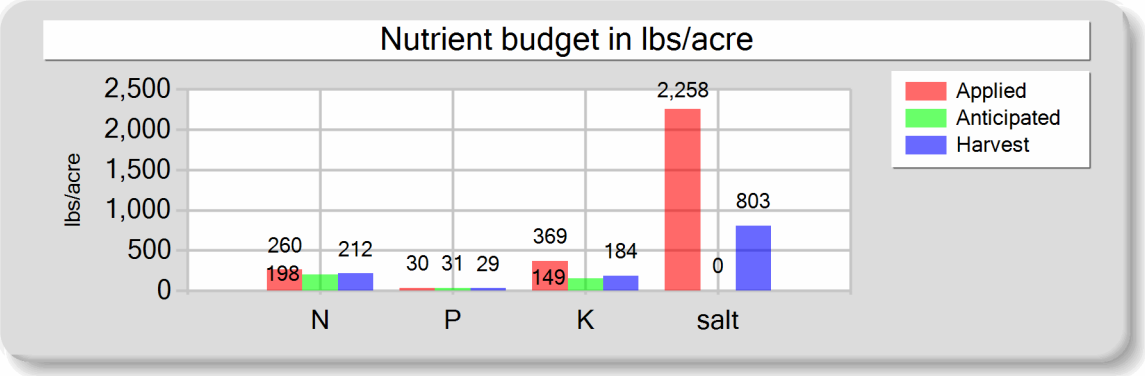
Fresh water applied	
53,317,240.00	*gallons*
1,963.49	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
1,050,000.00	*gallons*
38.67	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pavao - 09/27/2022: Wheat, silage, soft dough

Field name: Pavao | Crop: Wheat, silage, soft dough | Plant date: 09/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.26	30.20	369.50	2,257.98
Fresh water	29.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.01	30.20	369.50	2,257.98
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	211.89	29.00	184.01	802.97
Nutrient balance	48.12	1.20	185.49	1,455.01
Applied to removed ratio	1.23	1.04	2.01	2.81

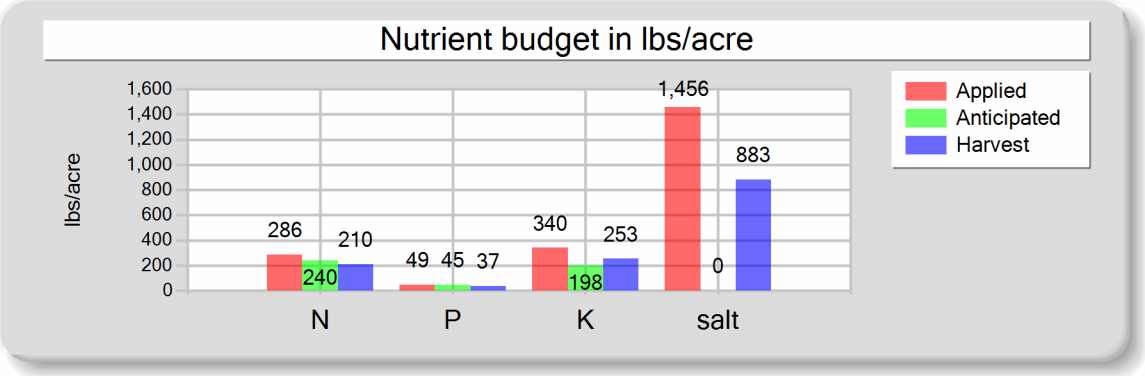
Fresh water applied	
34,830,000.00	*gallons*
1,282.67	*acre-inches*
15.09	*inches/acre*

Process wastewater applied	
5,460,000.00	*gallons*
201.07	*acre-inches*
2.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pavao - 05/01/2023: Corn, silage

Field name: Pavao | Crop: Corn, silage | Plant date: 05/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.21	48.64	339.93	1,456.39
Fresh water	76.56	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.77	48.64	339.93	1,456.39
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.92	36.65	253.23	882.98
Nutrient balance	75.85	11.98	86.70	573.41
Applied to removed ratio	1.36	1.33	1.34	1.65

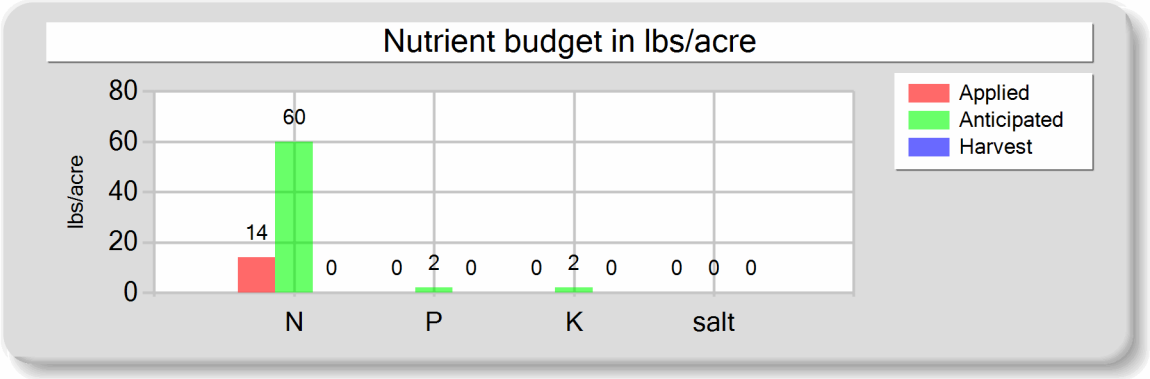
Fresh water applied	
118,890,000.00	*gallons*
4,378.31	*acre-inches*
51.51	*inches/acre*

Process wastewater applied	
4,830,000.00	*gallons*
177.87	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pavao pasture - 12/01/2014: Pasture

Field name: Pavao pasture | Crop: Pasture | Plant date: 12/01/2014



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	0.00
Anticipated crop nutrient removal	60.00	2.00	2.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	14.00	0.00	0.00	0.00
Applied to removed ratio	0.00	0.00	0.00	0.00

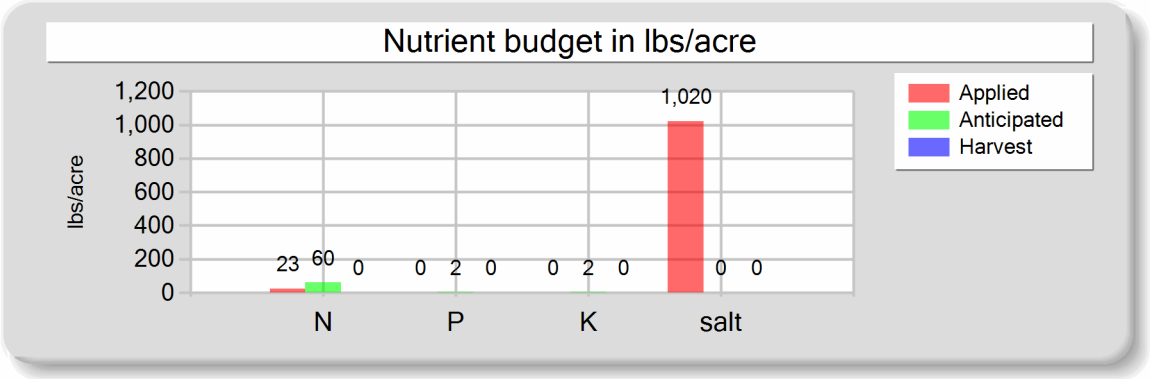
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Pereira - 11/30/2018: Pasture

Field name: Pereira | Crop: Pasture | Plant date: 11/30/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.67	0.00	0.00	1,020.41
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	22.67	0.00	0.00	1,020.41
Anticipated crop nutrient removal	60.00	2.00	2.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	22.67	0.00	0.00	1,020.41
Applied to removed ratio	0.00	0.00	0.00	0.00

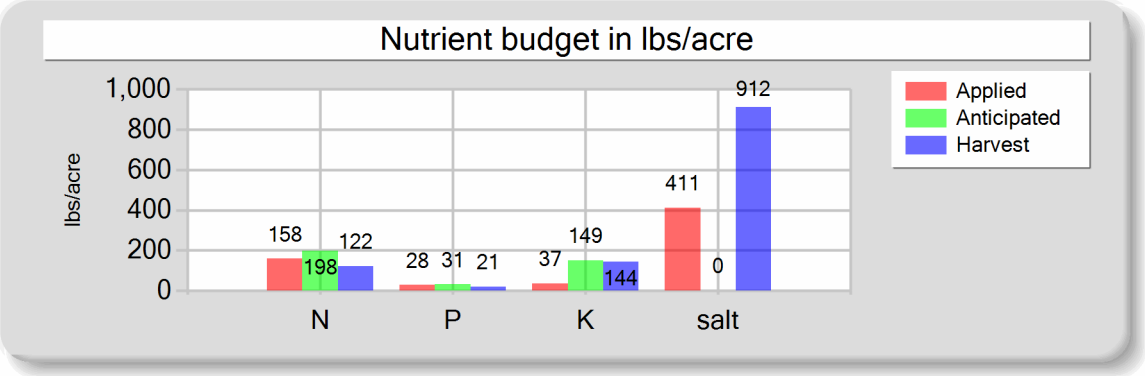
Fresh water applied	
45,854,130.00	*gallons*
1,688.65	*acre-inches*
22.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Vieira - 09/27/2022: Wheat, silage, soft dough

Field name: Vieira | Crop: Wheat, silage, soft dough | Plant date: 09/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	147.77	27.64	36.66	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.49	0.00	0.00	410.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.26	27.64	36.66	410.97
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	122.28	20.73	144.04	911.91
Nutrient balance	35.98	6.91	-107.38	-500.95
Applied to removed ratio	1.29	1.33	0.25	0.45

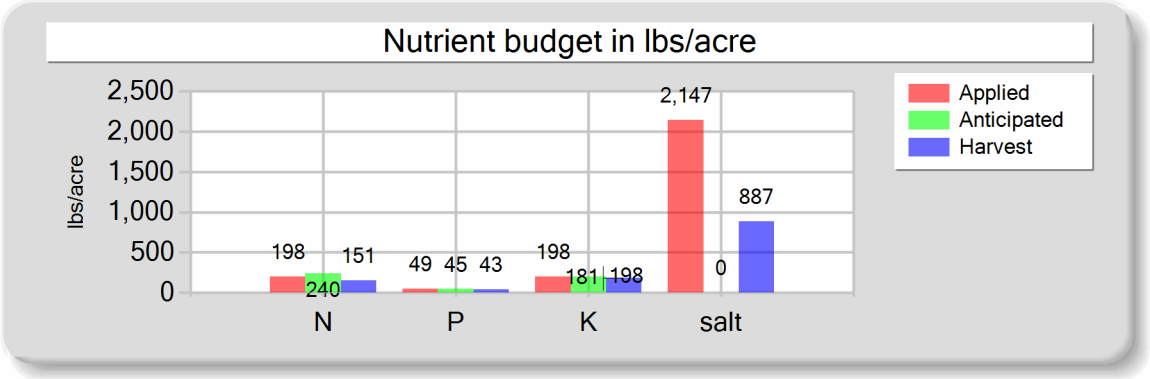
Fresh water applied	
15,512,840.00	*gallons*
571.29	*acre-inches*
9.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Vieira - 05/30/2023: Corn, silage

Field name: Vieira | Crop: Corn, silage | Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	172.39	49.25	197.96	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	18.25	0.00	0.00	2,147.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.64	49.25	197.96	2,147.22
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	151.16	42.72	180.73	887.24
Nutrient balance	46.48	6.54	17.23	1,259.98
Applied to removed ratio	1.31	1.15	1.10	2.42

Fresh water applied	
81,051,330.00	*gallons*
2,984.84	*acre-inches*
47.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Manure

Sample and source description: 2022 Fall Manure

Sample date: 10/05/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	4,900.00	6,500.00							
DL	1.00	1.00	1.00							

SSB Manure

Sample and source description: SSB Manure

Sample date: 10/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,500.00	4,200.00	8,600.00	18,200.00	5,800.00	3,200.00	6,700.00	4,000.00		28.60
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Compost Manure

Sample and source description: Compost Manure

Sample date: 12/13/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	5,200.00	20,900.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

2022 Q4 Lagoon

Sample and source description: 2022 Q4 Lagoon

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	338.00	235.00	0.00	1.90	54.70	717.00								8,530.00	4,240
DL	1.00	1.00	0.01	1.00	1.00	1.00								1.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	475.00	304.00	0.00	1.50	51.30	664.00								7,910.00	4,100
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	427.00	281.00	0.00	1.00	75.50	696.00								8,210.00	4,480
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	425.00	206.00	0.00	0.60	117.00	728.00								7,080.00	2,320
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	518.00	339.00	0.00	0.20	105.00	1,080.00								10,000.00	6,250
DL	1.00	1.00	0.01	0.01	1.00	1.00								1.00	1

C. FRESH WATER ANALYSES

CCID water

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.70	0.00	1.70								328.00	200
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW East

Sample description: DW East

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.10	0.00	5.10								1,200.00	
DL	1.00	0.01	1.00								1.00	

Domestic Well

DW Heifers

Sample description: DW Heifers

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.90	0.00	5.90								1,300.00	
DL	1.00	0.01	1.00								1.00	

DW West

Sample description: DW West

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.90	0.00	4.90								1,120.00	
DL	1.00	0.01	1.00								1.00	

Drain water

Drain Water

Sample description: Drain Water

Sample date: 06/01/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.80	0.00	0.80								653.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

IW Big

Sample description: IW Big

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.70	0.00	8.70								1,410.00	
DL	1.00	0.01	1.00								1.00	

IW Little

Sample description: IW Little

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.50	0.00	3.50								1,280.00	
DL	1.00	0.01	1.00								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1-1 - 09/25/2022: Ryegrass silage

CFD 1-1 Ryegrass

Sample and source description: CFD 1-1 Ryegrass

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	5,200.00	46,900.00		14.80
DL	0.01	0.01	0.01		1.00

1-2 - 09/27/2022: Wheat, silage, soft dough

CFD 1-2 Wheat Silage

Sample and source description: CFD 1-2 Wheat Silage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	2,200.00	14,400.00		8.70
DL	0.01	0.01	0.01		1.00

1-2 - 05/01/2023: Corn, silage

CFD 1-2 Corn Silage

Sample and source description: CFD 1-2 Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,100.00	2,100.00	24,500.00		7.70
DL	0.01	0.01	0.01		1.00

1-3 - 09/25/2022: Ryegrass Silage

CFD 1-3 Ryegrass

Sample and source description: CFD 1-3 Ryegrass

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	4,200.00	40,300.00		12.50
DL	0.01	0.01	0.01		1.00

1-3 - 09/25/2022: Ryegrass Silage

CFD 1-3 Ryegrass

Sample and source description: CFD 1-3 Ryegrass

Sample date: 05/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,700.00	5,000.00	43,000.00		13.70
DL	0.01	0.01	0.01		1.00

1-3 - 05/28/2023: Sudangrass, silage

CFD 1-3 Sudan Silage

Sample and source description: CFD 1-3 Sudan Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,300.00	3,900.00	38,900.00		17.80
DL	0.01	0.01	0.01		1.00

Frontella - 09/25/2022: Wheat, silage, soft dough

Frontella Wheat Silage

Sample and source description: Frontella Wheat Silage

Sample date: 05/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	2,400.00	15,900.00		6.80
DL	0.01	0.01	0.01		1.00

Frontella - 05/30/2023: Corn, silage

Frontella Corn Silage

Sample and source description: Frontella Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	1,800.00	9,800.00		3.90
DL	0.01	0.01	0.01		1.00

Pavao - 09/27/2022: Wheat, silage, soft dough

Pavao Wheat Silage

Sample and source description: Pavao Wheat Silage

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,000.00	2,600.00	16,500.00		7.20
DL	0.01	0.01	0.01		1.00

Pavao - 05/01/2023: Corn, silage

Pavao Corn Silage

Sample and source description: Pavao Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,200.00	15,200.00		5.30
DL	0.01	0.01	0.01		1.00

Vieira - 09/27/2022: Wheat, silage, soft dough

Vieira Wheat Silage

Sample and source description: Vieira Wheat Silage

Sample date: 05/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,000.00	13,900.00		8.80
DL	0.01	0.01	0.01		1.00

Vieira - 05/30/2023: Corn, silage

Vieira Corn Silage

Sample and source description: Vieira Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,200.00	2,600.00	11,000.00		5.40
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

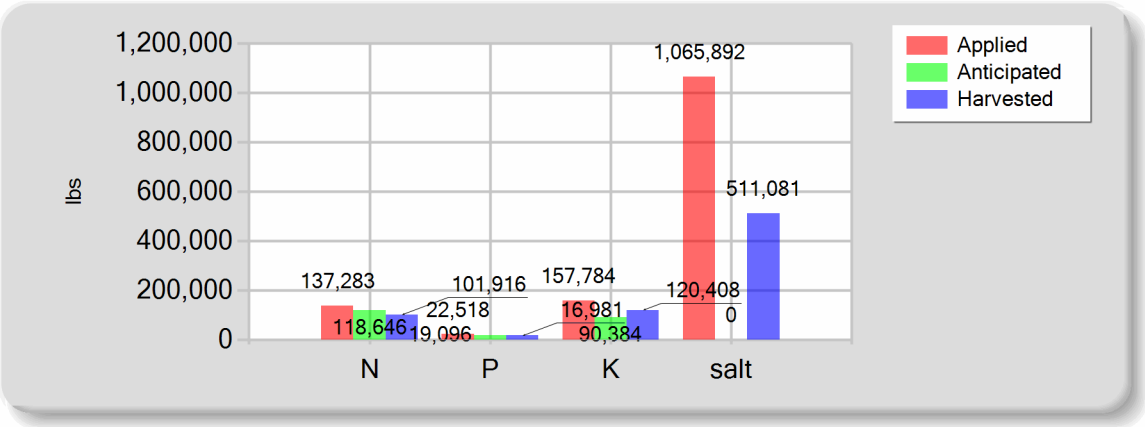
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

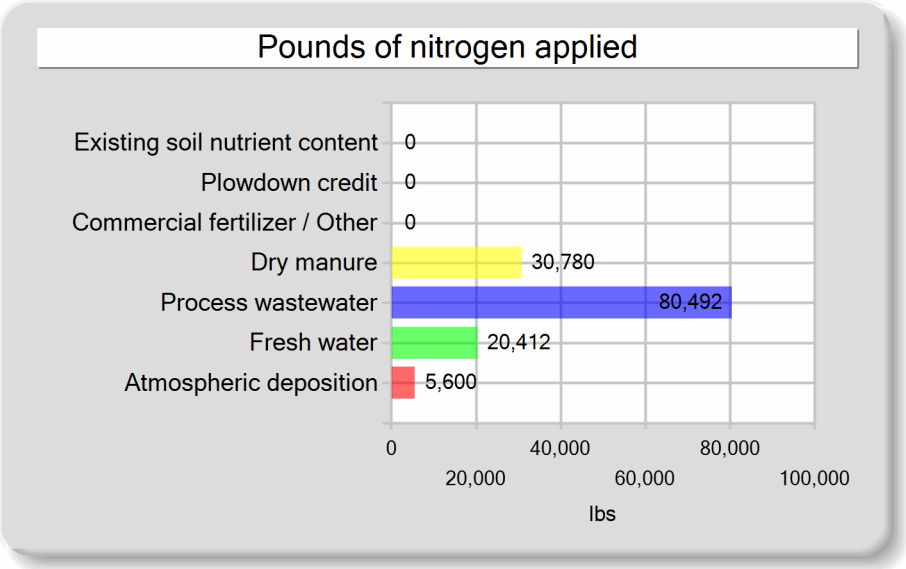
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

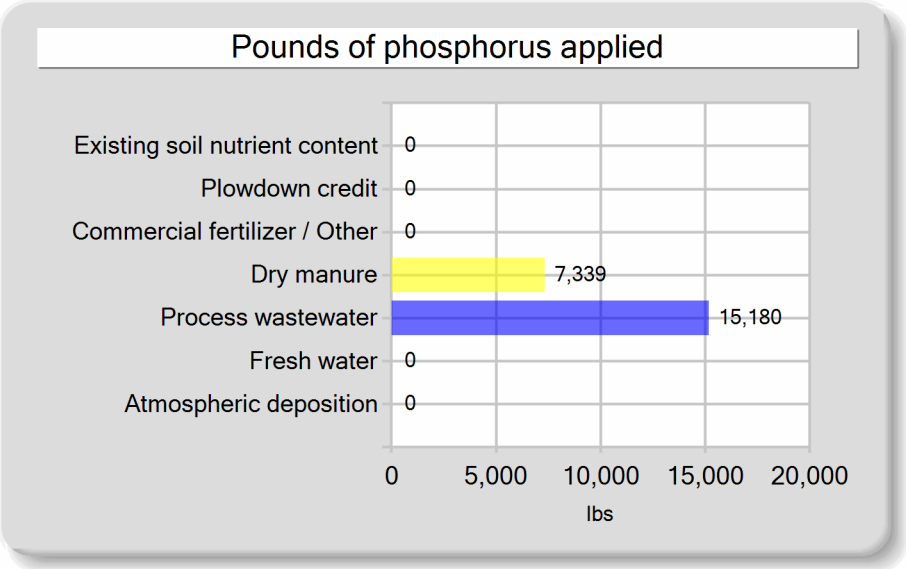
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	30,779.55	7,338.73	22,069.39	0.00
Process wastewater	80,492.13	15,179.55	135,714.11	709,619.58
Fresh water	20,411.59	0.00	0.00	356,272.25
Atmospheric deposition	5,600.00	0.00	0.00	0.00
Total nutrients applied	137,283.28	22,518.28	157,783.50	1,065,891.83
Anticipated crop nutrient removal	118,646.00	19,096.20	90,383.80	0.00
Actual crop nutrient removal	101,916.04	16,981.30	120,408.23	511,080.82
Nutrient balance	35,367.24	5,536.98	37,375.28	554,811.01
Applied to removed ratio	1.35	1.33	1.31	2.09

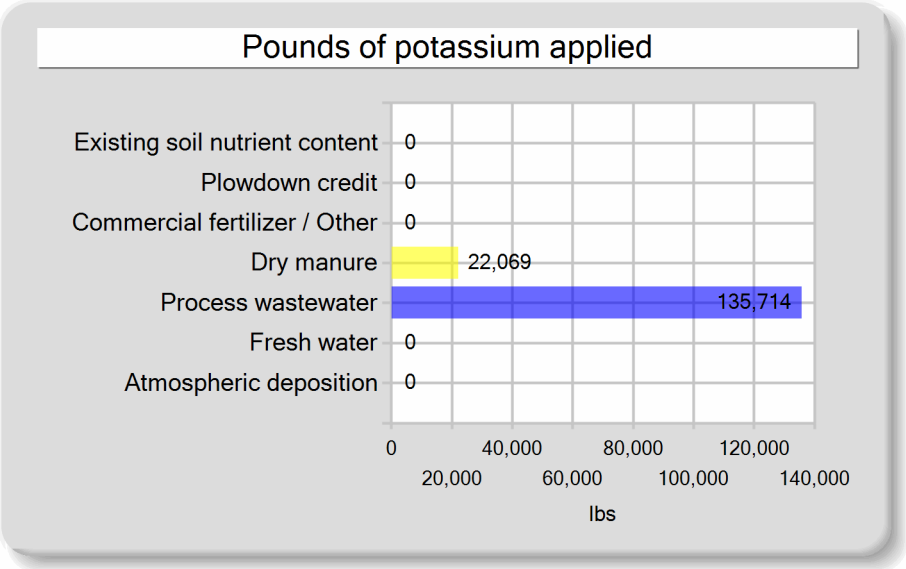
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

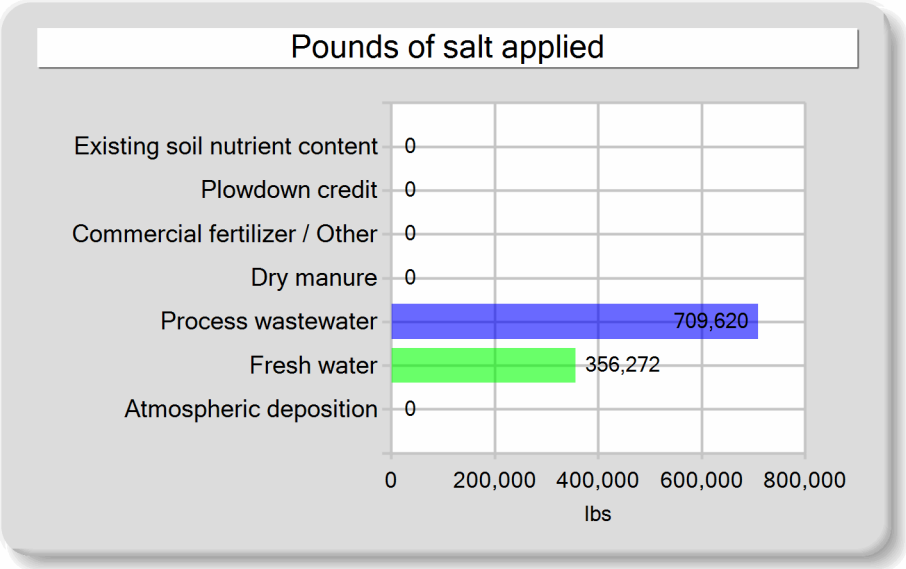


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Pavao pasture field dry farmed therefore no irrigation or harvest events submitted.

ANNUAL REPORT VALIDATION INFORMATION

A. VALIDATION ERRORS

The following sections contain validation errors and should be reviewed before submitting the Annual Report :

1. Harvest Events
2. Land Application Events

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ezequiel Correia

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ezequiel Correia	Ezquiel Correia, Jr
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 28, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Junior Correia

Name of Dairy Facility: Correia Family Dairy

Facility Address:

26380 W Fahey RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Junior Correia (209) 652-0029
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J & F Fertilizer

Address of Hauling Company/Person:

1275 North AVE	Gustine	CA	95322
Number and Street	City	State	Zip Code

Contact Person: John Nunes (209) 495-1964
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Broker

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

J & F Fertilizer (209) 495-1964
Name Phone Number

1275 North AVE	Gustine	CA	95322
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

	Gustine	95322
Address	City	Zip Code

Meredith Ave	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 05/20/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 5,096.00 tons

Manure Solids Content: 59.2 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Ezequiel Correia — 5/22/23

Operator Signature — Date

John Nunes — 5/22/23

Hauler Signature — Date

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Correia Family Dairy - Gustine
26380 W Fahey Road
Gustine CA 95322
00-0021367 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1001
Sampled By: M DeKeyrel
Requested By: Junior Correia
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	IW - Big	12/13/23	13:10	1410	1.41	8.7	ND		7.6
2	IW - Little	12/13/23	13:11	1280	1.28	3.5	ND		7.7
3	DW - West	12/13/23	13:12	1120	1.12	4.9	ND		7.7
4	DW - East	12/13/23	13:13	1200	1.20	5.1	ND		7.8
5	DW - Heifers	12/13/23	13:15	1300	1.30	5.9	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

12/14/23 14:20 23L1001

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21367 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Correia Family Dairy
Address: 26380 W Fahey Road
City, State, Zip: Gustine, CA 95322
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Junior Correia - (209) 652-0029

Ranch _____

Date sampled: 12/13/23

Sampled by: M. Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 5 No. Bottles: 5

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. WW-Big	Sampled From:	12/13	1:10	Ø	1.8
2. WW-Little	Sampled From:	12/13	1:11	Ø	2.9
3. DW-West	Sampled From:	12/13	1:12	Ø	2.3
4. DW-East	Sampled From:	12/13	1:13	Ø	3.0
5. DW-Heifers	Sampled From:	12/13	1:15	Ø	4.1
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeyrel	CAS Inc	12/13/23 @ 1:10	12/14/23 1:10
Second	JLF	DLV	12-14-23 12:00	12-14-23 2:15
Third				
Fourth		DLV	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1	1	1					
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L1001

SEND OUT